

Merging and aggregating data

CASE STUDY: FINANCIAL ANALYSIS IN KNIME

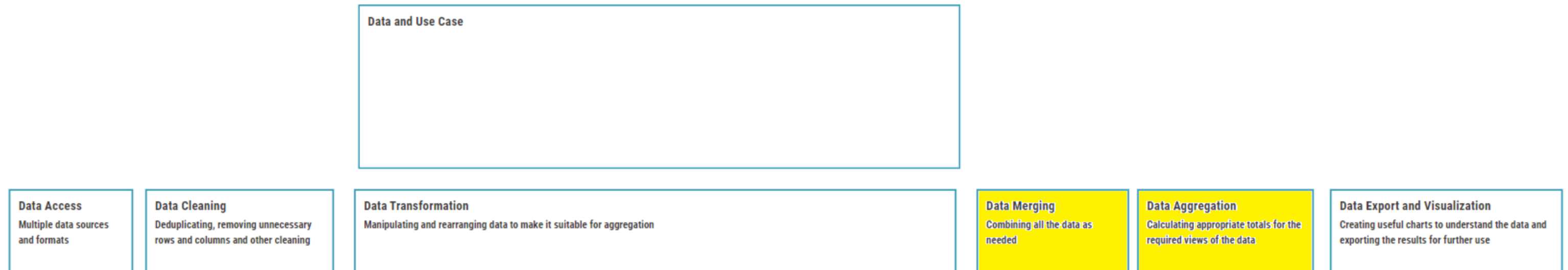


Andrew Logan

Data Scientist, 4&8 Ltd

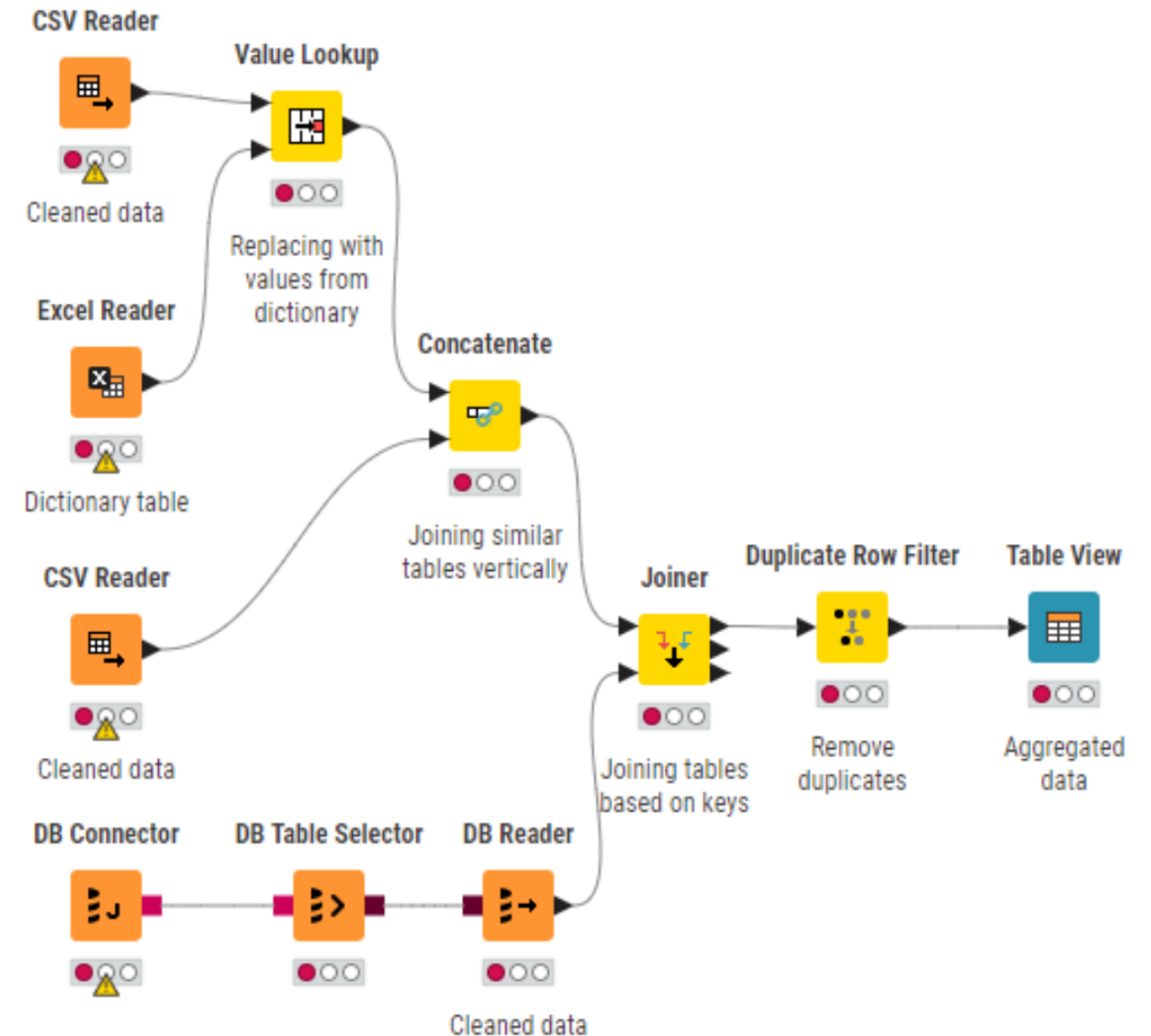
Back to the plan

- Cleaned and transformed data
- Merging: combining into one consistent table
- Aggregating: summarizing data as defined in the business case



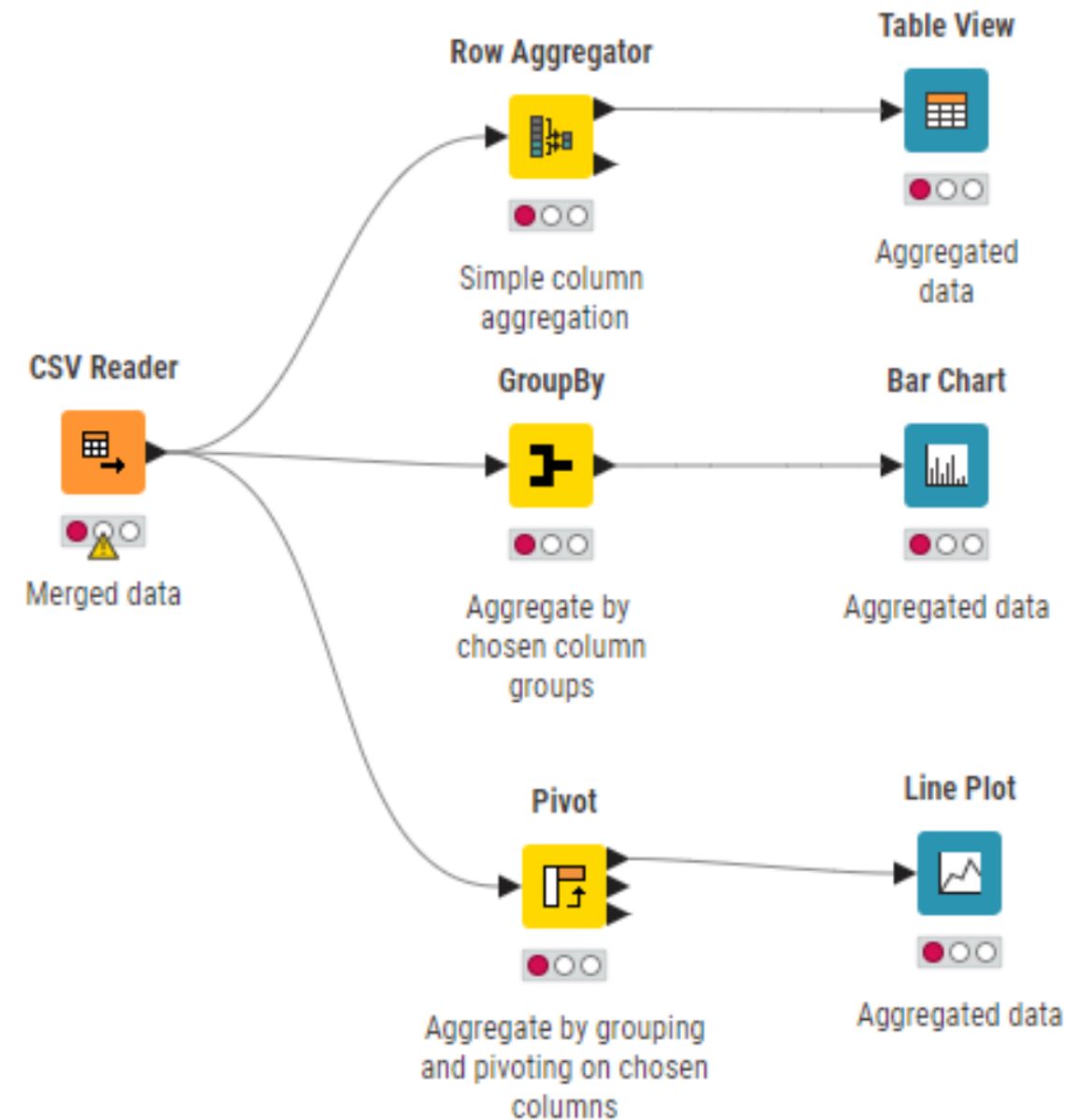
All in one

- Most common nodes:
 - Value Lookup, Concatenate, Joiner
- Deduplicate
- Annotate!



Summarizing it all

- Most common nodes:
 - Row Aggregator, GroupBy, Pivot
- Keep it simple and check



Using the right aggregations

- Business case

Using the right aggregations

- Business case
- Row Aggregator

Row Aggregation

Sum quantity

title	genre	audience	quantity
book1	adventure	teen	2
book2	adventure	adult	2
book3	fantasy	teen	3
book4	fantasy	adult	2
book5	romance	adult	2
book6	adventure	teen	1
book7	adventure	adult	2
book8	fantasy	teen	2
book9	fantasy	adult	1
book10	romance	adult	1

total
18

Using the right aggregations

- Business case
- Row Aggregator
- **Groupby**

Groupby

Group by genre (rows), Sum quantity

title	genre	audience	quantity
book1	adventure	teen	2
book2	adventure	adult	2
book3	fantasy	teen	3
book4	fantasy	adult	2
book5	romance	adult	2
book6	adventure	teen	1
book7	adventure	adult	2
book8	fantasy	teen	2
book9	fantasy	adult	1
book10	romance	adult	1

genre	total
adventure	7
fantasy	8
romance	3

Using the right aggregations

- Business case
- Row Aggregator
- Groupby
- **Pivot**

Pivot

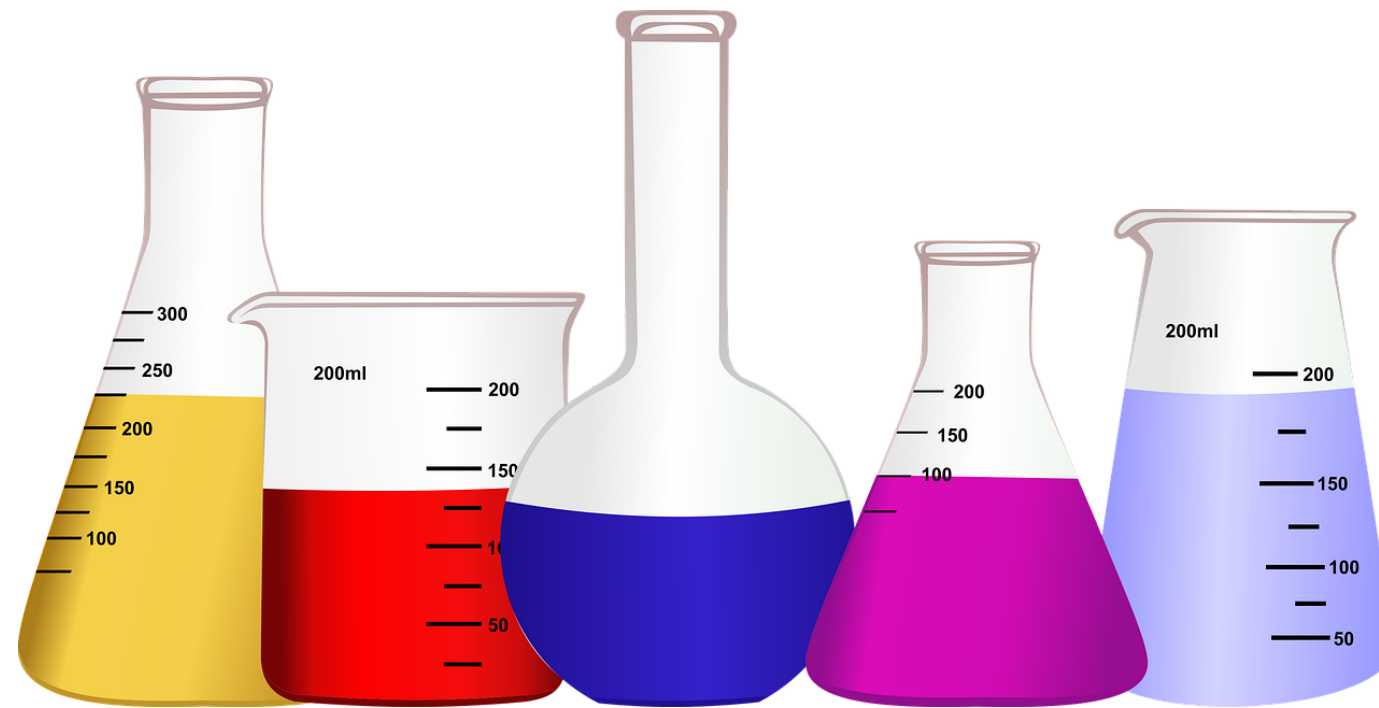
Group by genre (rows), Pivot by audience (rows to columns), Sum quantity

title	genre	audience	quantity
book1	adventure	teen	2
book2	adventure	adult	2
book3	fantasy	teen	3
book4	fantasy	adult	2
book5	romance	adult	2
book6	adventure	teen	1
book7	adventure	adult	2
book8	fantasy	teen	2
book9	fantasy	adult	1
book10	romance	adult	1

genre	adult total	teen total
adventure	4	3
fantasy	3	5
romance	3	

Using the right aggregations

- Business case
- Row Aggregator
- Groupby
- Pivot
- Experiment



Let's practice!

CASE STUDY: FINANCIAL ANALYSIS IN KNIME